SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **G.Mohammad shareef/(192311288)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Mohammad shareef

RUBRICS FOR CALCULATION:

Numerical Problems						
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterpret s or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem
Solving

1. Data Cleaning using Duplicate Detection (Patient Registration) (5 Marks) The following patient IDs were extracted from a registration database.
P101, P102, P105, P103, P105, P107, P108, P103, P110, P112
Perform the following:

1. **Identify duplicate records.**
2. **Remove duplicate entries.**
3. **Determine the number of unique patients.**

2.Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)
The following data represent treatment cost (₹ thousand) and hospital stay (days).

Treatment Cost (x)	**Hospital Stay (y)**
20	**3**
25	**4**
30	**5**
35	**6**
40	**8**

Perform the following:

1. **Compute the mean of x and y.**
2. **Calculate the covariance.**
3. **Interpret the relationship between treatment cost and duration of stay.**

3.Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks) Recovery time (days) for patients after surgery is recorded as:
8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35
Perform the following:

1. **Compute the mean.**
2. **Compute the standard deviation.**
3. **Calculate the z-score for each observation.**
4. **Identify the outlier(s) using the criterion $|z| > 3$.**

4.Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks) A clinic records fasting blood glucose levels (mg/dL).
Sorted Data:

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145
Partition the data into 3 equal-width bins.
Replace each observation by the nearest bin boundary. Write the transformed dataset.

5.Standardization of Insurance Claim Amounts (10 Marks) An insurance company records claim amounts (₹ thousand): 12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Perform the following:

1. **Compute the mean.**
2. **Compute the standard deviation.**
3. **Calculate the z-score for every observation.**
4. **Identify claims that are above one standard deviation from the mean.**

Answers

1. Data Cleaning using Duplicate Detection (Patient Registration)
Sol:Given Patient IDs:
P101, P102, P105, P103, P105, P107, P108, P103, P110, P112
Step 1: Identify Duplicate Records
The duplicate patient IDs are:

- P105
- P103

Step 2: Remove Duplicate Entries
After removing duplicates, the unique patient IDs are:
P101, P102, P105, P103, P107, P108, P110, P112
Step 3: Determine the Number of Unique Patients
Total unique patient IDs:8

Final Answer

- Duplicate Records: P105, P103
- Dataset after Removing Duplicates: P101, P102, P105, P103, P107, P108, P110, P112
- Number of Unique Patients: 8

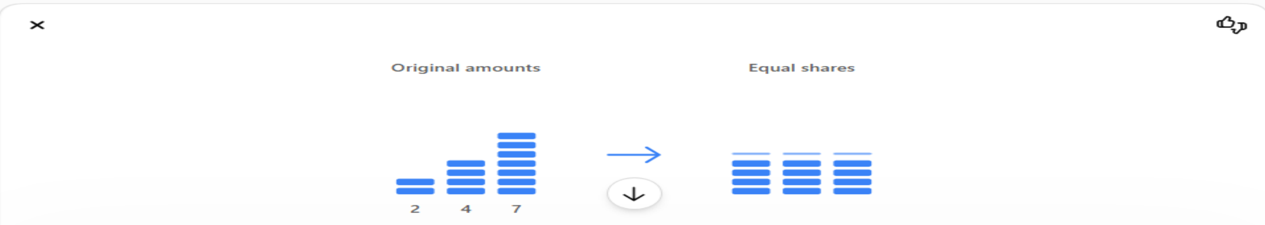
2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)
The following data represent treatment cost (₹ thousand) and hospital stay (days).

Sol:

2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)

Given Data

Treatment Cost (x) (₹ thousand)	Hospital Stay (y) (days)
20	3
25	4
30	5
35	6
40	8



1.

Step 1: Compute the Mean of x and y

Mean of Treatment Cost (x):

$$\bar{x} = \frac{20 + 25 + 30 + 35 + 40}{5} = \frac{150}{5} = 30$$

Mean of x = 30 (₹ thousand)

Mean of Hospital Stay (y):

$$\bar{y} = \frac{3 + 4 + 5 + 6 + 8}{5} = \frac{26}{5} = 5.2$$

Mean of y = 5.2 days

Step 2: Calculate the Covariance

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
20	3	−10	−2.2	22
25	4	−5	−1.2	6
30	5	0	−0.2	0
35	6	5	0.8	4
40	8	10	2.8	28

Sum of products = 22 + 6 + 0 + 4 + 28 = 60

Covariance (using $n = 5$):

$$\text{Cov}(x, y) = \frac{60}{5} = 12$$

Covariance = 12

Step 3: Interpretation

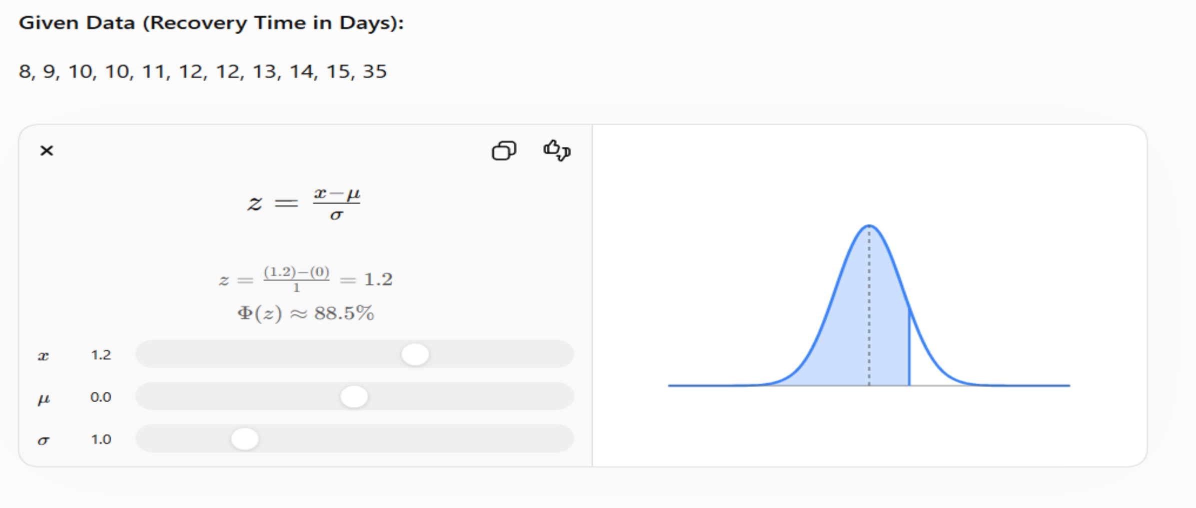
Since the covariance is positive (12), there is a positive relationship between treatment cost and hospital stay.

Interpretation: As the treatment cost increases, the hospital stay tends to increase.

3. Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks)

Recovery time (days) for patients after surgery is recorded as:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35



Sol:

1.

Sum of observations:

8 + 9 + 10 + 10 + 11 + 12 + 12 + 13 + 14 + 15 + 35 = 149

Number of observations (n) = 11

$$\text{Mean} = \frac{149}{11} = 13.55$$

Mean = 13.55 days

Step 2: Compute the Standard Deviation

Using the population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$\sigma \approx 7.25$$

Standard Deviation = 7.25 days

Step 3: Calculate the Z-Score

$$\boldsymbol{z = \frac{x - \overline{x}}{\sigma}}$$

Recovery Time (x)	**Z-Score**
8	**-0.77**
9	**-0.63**
10	**-0.49**
10	**-0.49**

Recovery Time (x)	Z-Score
11	**-0.35**
12	**-0.21**
12	**-0.21**
13	**-0.08**
14	**0.06**
15	**0.20**

Step 4: Identify the Outlier(s)
Criterion:

$$|z| > 3$$

The largest z-score is:

- 35 → z = 2.96

Since 2.96 < 3, there are no outliers according to the given criterion.

4. Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks)

Sol: Given Sorted Data (mg/dL):
75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

Step 1: Calculate Bin Width

- Minimum value = 75
- Maximum value = 145
- Number of bins = 3

$$\text{Bin Width} = \frac{145 - 75}{3} = \frac{70}{3} \approx 23.33$$

Step 2: Form Equal-Width Bins

- Bin 1: 75 – 98.33
- Bin 2: 98.34 – 121.66
- Bin 3: 121.67 – 145

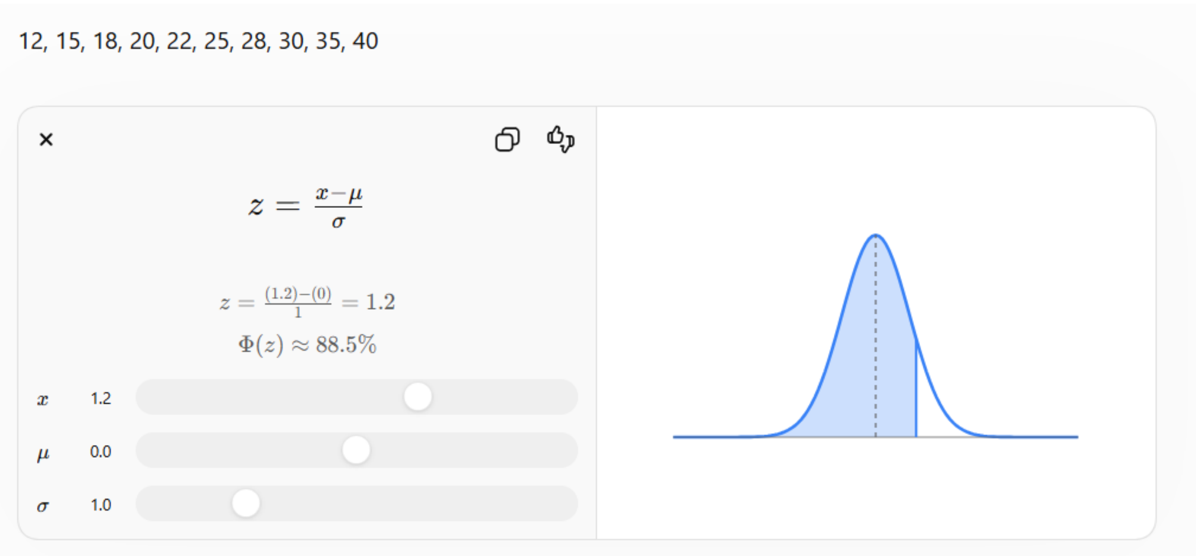
Step 3: Replace Each Observation by the Nearest Bin Boundary

Original Value	Nearest Boundary	Replaced Value
75	**75**	**75**
82	**75**	**75**
88	**98.33**	**98.33**
92	**98.33**	**98.33**

Original Value	Nearest Boundary	Replaced Value
95	98.33	98.33
99	98.34	98.34
104	98.34	98.34
110	121.66	121.66
118	121.66	121.66
122	121.67	121.67
130	121.67	121.67
145	145	145

Step 4: Transformed Dataset
75, 75, 98.33, 98.33, 98.33, 98.34, 98.34, 121.66, 121.66, 121.67, 121.67, 145

5. Standardization of Insurance Claim Amounts (10 Marks)
An insurance company records claim amounts (₹ thousand):
12, 15, 18, 20, 22, 25, 28, 30, 35, 40
Perform the following:
Sol: Given Claim Amounts (₹ thousand):



Step 1: Compute the Mean

Sum of observations:
12 + 15 + 18 + 20 + 22 + 25 + 28 + 30 + 35 + 40 = 245
Number of observations (n) = 10

$$\text{Mean} = \frac{245}{10} = 24.5$$

Mean = 24.5 (₹ thousand)
Step 2: Compute the Standard Deviation
Using the population standard deviation formula:

$$\sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

$$\boldsymbol{\sigma \approx 8.20}$$

Standard Deviation = 8.20 (₹ thousand)
Step 3: Calculate the Z-Score

$$\boldsymbol{z = \frac{x - \bar{x}}{\sigma}}$$

Claim Amount (₹ thousand)	**Z-Score**
12	**-1.52**
15	**-1.16**
18	**-0.79**
20	**-0.55**
22	**-0.30**
25	**0.06**
28	**0.43**
30	**0.67**
35	**1.28**
40	**1.89**

Step 4: Identify Claims Above One Standard Deviation
Criterion:

$$z > 1$$

The claim amounts with z-score > 1 are:

- ₹35 thousand (z = 1.28)
- ₹40 thousand (z = 1.89)